Applications Development Project 3 –ADPB301

Software Requirements Specification (SRS)

I have read, understood and agree to the contents of this document. I accept full responsibility for any errors and omissions in this document.

Group number: _____ Group name: _____

	Student No	First name	Surname	Sign	Meetings Held	No attended
1						
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Business Domain	
Business name	
Contact Person	
Contact number	
Email address	

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1. Executive summary

MuhleOps is a reliable online system designed to meet the operational and financial management requirements of Muhle Building. The system aims to centralize and digitize essential business processes such as recruitment, client onboarding, project planning, resource allocation, equipment maintenance, construction execution, progress monitoring, financial management, and project handover. The platform offers immediate insight into project performance and cost management across various construction sites by substituting manual spreadsheets and disjoined communication methods. The system supports multiple user roles including Admin, Client, Project Manager, Site Manager and Finance Officer. Each role has specific dashboards and functionalities tailored to their responsibilities in the construction project life cycle.

2. Business processes (List all Business processes for increment 2)

No	Business process	Sem 2 / Inc	Brief description
1.	Recruit & Hire Staff	2/1	Manages job postings, applications, interviews and onboarding of new employees.
2.	Onboard New Client	2/1	Register new client companies, create contracts and manage client signatures.
3.	Plan & Scope Project	2/1	Define project scope, objectives, deliverables and estimated costs.
4.	Allocate Resources & Staff	2/1	Assign internal staff and equipment to active projects.
5.	Execute Construction Work	2/1	Carry out on-site construction activities according to schedule and quality standards.
6.	Maintain Equipment & Assets	2/2	Schedule and track maintenance, repairs and inspections of equipment.
7.	Monitor Progress & Quality	2/2	Track project progress, conduct inspections and manage variations.
8.	Manage Project Finance & Payroll	2/2	Approve timesheets, process payroll and generate client invoices.
9.	Close Project & Handover	2/2	Complete final inspections, obtain client sign-off, and hand over the project.

3. Business use cases (List all Business use cases in sequence for semester 2, increment 1 and 2)

No	Business process	Use case	Sem / Inc	Brief Use Case description	Full Name	Completed Yes / No
1.	Recruit & Hire Staff	Create Vacancy	2/1	Admin creates and publishes job vacancy on the system.	Vukile	Yes

2.	Recruit & Hire Staff	Job application	2/1	Candidates submit job application for published positions.	Sihawukele	Yes
3.	Recruit & Hire Staff	Shortlist Applications	2/1	Admin shortlists qualified candidates.	Dlomo	Yes
4.	Recruit & Hire Staff	Schedule interview	2/1	Admin schedules interviews with shortlisted candidates.	Thabiso	Yes
5.	Recruit & Hire Staff	Hire Staff	2/1	Admin selects a candidate, create staff profile and sends welcome email.	Sbonelo	Yes
6.	Onboard New Client	Register Client	2/1	Admin registers new client companies and creates client profiles.	Minenhle	Yes
7	Onboard New Client	Create contract	2/1	Project Manager creates and sends service contracts to clients	Mqadi	Yes
8	Onboard New Client	Sign contract	2/1	Client signs the contract and upload signed copy.	Thandolwethu	Yes
9	Plan & Scope Project	Initiate project	2/1	Project Manager defines scope, objectives, deliverables and estimated costs.	Nhlakanipho	Yes
10	Allocate Resources & Staff	Assign staff	2/1	Project Manager assigns staff to specific project roles	Mnwana	Yes
11	Allocate Resources & Staff	Assign equipment.	2/1	Project Manager assign equipment to the project.		
12	Execute Construction Work	Assign daily tasks	2/1	Project Manager assigns daily tasks to crew members.		
13	Execute Construction Work	Assign Equipment	2/2	Project Manager assigns equipment to the project or crew		
14	Monitor Progress & Quality	Clock In	2/2	Crew member clocks in to indicate commencement of work		
15	Maintain Equipment & Assets	Log Damaged Equipment	2/2	Crew member logs the details of damaged equipment encountered.		
16	Monitor Progress & Quality	Update Task Status	2/2	Crew member updates the progress status of the task		
17	Monitor Progress & Quality	Clock Out	2/2	Crew member clocks out to indicate conclusion of work.		

18	Maintain Equipment & Assets	Checkin Damaged Equipment	2/2	Admin checks in equipment for repair.		
19	Maintain Equipment & Assets	Checkout Fixed Equipment	2/2	Admin checks out fixed equipment for use.		
20	Monitor Progress & Quality	Request Variation	2/2	Client request for project variation		
21	Monitor Progress & Quality	Approve Variation	2/2	Project Manager approves project variation		
22	Manage Project Finance & Payroll	Process Payroll	2/2	Finance Officer processes payroll.		
23	Manage Project Finance & Payroll	Close Project	2/2	Project Manager closes the project upon completion.		

4. Use case description *(Only describe Semester 2 Increment 2 use cases)*

4.1 Use case description

Use case number	12
Use case name	Assign Tasks
Brief description	Project Manager assigns tasks to the staff.
Triggering event	Project allocation needs
Actor(s)	Project Manager
Preconditions	Project and staff are active
Flow of activities	1. Project Manager navigates to the task allocation page. 2. System displays tasks to be assigned. 3. System displays list of all available staff. 4. Project Manager selects crew members. 5. Project Manager confirms the allocation. 6. System notifies the assigned staff of the task.

Use case number	13
Use case name	Assign Equipment
Brief description	Project Manager assigns equipment to the project or crew.
Triggering event	Equipment allocation needs.
Actor(s)	Project Manager
Preconditions	Equipment is registered and available
Flow of activities	1. Project Manager navigates to the equipment view.

	2. System displays the available equipment. 3. Project Manager selects equipment to be allocated and the project or staff. 4. Project Manager approves the allocation. 5. System updates equipment status. 6. System notifies all stakeholders.

Use case number	14
Use case name	Clock In
Brief description	Crew members clocks in to work.
Triggering event	Beginning of shift.
Actor(s)	Crew Member
Preconditions	Crew member is registered and assigned to a project for the day.
Flow of activities	1. Crew member logs into the system. 2. Crew member scans the QR Code displayed on the gate. 3. System confirms shift details against project schedule. 4. System records records time and location of the crew member. 5. System displays a confirmation message to the crew member.

Use case number	15
Use case name	Log damaged equipment
Brief description	Crew member logs details of damaged equipments.
Triggering event	Discovery of damaged equipment
Actor(s)	Crew Member
Preconditions	Equipment is registered in the system and crew member is logged in.
Flow of activities	1. Crew member access the equipment log. 2. Crew member selects the damaged equipment. 3. Crew member submits the damage report. 4. System marks equipment damaged 5. System notifies the Admin about the damage

Use case number	16
Use case name	Update Task status
Brief description	Site Manager updated the progress status of the task.
Triggering event	Completion of work activities
Actor(s)	Site Manager
Preconditions	Task has been assigned
Flow of activities	1. Site Manager access assigned tasks 2. Site Manager updates progress status and attach evidence. 3. System timestamps the update and updates the overall project progress. 4. System notifies Project Manager and Admin

Use case number	17
Use case name	Check Out
Brief description	Crew member clocks out to indicate conclusion of work
Triggering event	End of the shift for crew member
Actor(s)	Crew member
Preconditions	Crew member has clocked in for the day
Flow of activities	1. Crew member access the application and selects Clock-Out. 2. System records the time and hours worked. 3. System displays a summary of today's work. 4. System notifies crew member.

Use case number	18
Use case name	CheckIn Equipment
Brief description	Admin checks in equipment for repair
Triggering event	Equipment has been reported damaged or needs servicing
Actor(s)	Admin
Preconditions	Equipment is registered.
Flow of activities	1. Admin access the Equiment Maintenance portal. 2. Admin logs the check-in details and timelines. 3. The system updates the equipment status to In Repair. 4. System generates a repair tracking ticket. 5. System logs the check-in event

Use case number	19
Use case name	Checkout Fixed Equipment
Brief description	Admin checks out fixed equipment for use.
Triggering event	Equipment has been repaired and is ready for use.
Actor(s)	Admin
Preconditions	The equipment status is In Repair.
Flow of activities	1. Admin access the equipment repair records. 2. Admin selects the equipment that has been repaired. 3. Admin logs the repair checkout details and technical notes. 4. Admin updated the equipment status to indicate the equipment is available for use. 5. System clears the repair ticket 6. System notifies the Project Manager and Site Manager about equipment availability.

Use case number	20
Use case name	Request Project Variation
Brief description	Client request for project variations.
Triggering event	Project variations are identified and requested.

Actor(s)	Client or Project Manager
Preconditions	Project is active.
Flow of activities	1. Client accesses the project variation portal. 2. Client logs requested variation details. 3. Client submits formal request for project variation. 4. System generates a project variation reference number. 5. System notifies the Project

Use case number	21
Use case name	Approve Variation
Brief description	Project Manager approves project variations.
Triggering event	The request for project variation has been submitted.
Actor(s)	Project Manager
Preconditions	A request for variation has been submitted
Flow of activities	1. Project Manager accesses the list of submitted project variations. 2. Project Manager reviews the variation requests and their impact. 3. Project Manager approves or rejects the request for project variation. 4. If the variation is approved, the Project Manager logs the implementation details. 5. System updates the project with the approved variation and generates a formal variation order. 6. System notifies the client about the variation approval.

Use case number	22
Use case name	Process Payroll
Brief description	Finance Officer processes monthly payroll and initiate payments for all crew members.
Triggering event	End of monthly pay cycle.
Actor(s)	Finance Officer
Preconditions	All crew members have submitted their timesheets for the pay period
Flow of activities	1. Finance Officer accesses the Payroll portal. 2. The system compiles the approved timesheets to generate a payroll summary. 3. The system computes the amount due based on the approved hours. 4. Finance Officer reviews the generated payroll report. 5. Finance Officer approves the payroll process. 6. The system records the payroll process and generates electronic payslips.

Use case number	23
Use case name	Close Project
Brief description	Project Manager closes the project upon completion
Triggering event	The project has been physically completed.
Actor(s)	Project Manager
Preconditions	All project activities has been finalized.
Flow of activities	1. The relevant user accesses the project closure portal. 2. The user confirms the completion of all activities and closes the project. 3. System performs a final validation of the project completion 4. System updates the project status to closed and restricts further modifications. 5. System archives the project files and notifies all stakeholders of project closure.

4.2 Use interfaces

4.3 Use case diagrams